

Guneev Pannu

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EDUCATION

University of Toronto – 3.96 GPA

2024 – 2028

HBSc Major in Computer Science and Statistics, Math Minor

Relevant Courses: Object-Oriented Development, Data Structures & Algorithms, Computer Organization, Software Design, Computer Theory, Unix, Systems Programming, Databases, Parallel Computing, Statistics and Probability

Accolades: Dean's List, Volunteer Note Taker

EXPERIENCE

Fullstack Developer

September 2025 - Present

UofT Machine Learning Club 🌐

- Developed and launched **11** web pages to publicize the talent of over **400** members with **Next.js** and **React**
- Integrated **Supabase** backend to update frontend team information reducing manual content updates by **80%**
- Revamped website design elements with new **Figma** based designs using **Shadcn** and **Tailwind CSS**
- Designed flexible and reusable React components saving development time when creating new site pages by **30%**

Facilitated Study Group Leader

January 2025 – April 2025

UofT Robert Gillespie Academic Skills Center 🌐

- Planned, advertised, and hosted facilitated study sessions for students taking **CSC148: Intro to Computer Science**
- Utilized supplemental teaching techniques to effectively guide over **20** students at once with planned lesson plans
- Applied joint feedback from superiors and **partners** on teaching techniques, and lesson plans to keep improving

PROJECTS

🌐 Blog Website Backend | *Spring Boot, Java, Maven, Git, PostgreSQL, Docker*

- Built a backend software application in a **week** that could support a blog post social media app, like Tumblr
- Designed relational **PostgreSQL** schema and integrated **Spring Data JPA** for efficient query handling
- Implemented **REST** apis that supported CRUD operations which were **verified** with **Postman**
- Containerized application with **Docker** to enable consistent local and production environments
- Implemented **JWT**-based authentication and role-based authorization using **Spring Security**
- Reduced boilerplate code by leveraging **Lombok** and **MapStruct**, improving development velocity

Paint Program | *Java, JavaFX, Maven, Git*

- Built a paint program that allowed users to draw many shapes and lines on a resizable canvas using **JavaFX**
- Applied the **MVC** design pattern that allowed for rapid development and collaboration using **Git**
- Engineered a system to save and load files of the current canvas using **Regex** and FSMs
- Collaborated extensively to complete the project in a week sprint, handling many pull requests and bug fixes

Custom Unix Shell | *C, Make, Git*

- Developed a custom **Unix** shell in **C** replicating core Bash functionality, including process creation, piping, and command parsing
- Implemented inter-process communication and socket-based client-server chat system over local networks
- Debugged and memory-optimized system using GDB and Valgrind, achieving zero memory leaks

🌐 Python Syntax Checker | *PyTorch, Git*

- Trained a machine learning model to detect simple Python syntax errors with **92.5%** test accuracy
- Created and labeled a **5,000 sample dataset** using Python's ast and tokenizer modules

TECHNICAL SKILLS

Languages: Java, JavaScript, Typescript, Python, C, RISC-V Assembly, LaTeX

Frameworks: Next.js, React, Spring Boot, Agile, Maven,

Developer Tools: Git, VSCode, IntelliJ IDEA, Docker, Postman, PyCharm, Netlify, WSL

Libraries: PyTorch, cProfile, NumPy, Matplotlib, Tailwind, OpenCV, PyAutoGUI